EYE COLOUR, INHERITANCE AND PROBABILITY ON ONE PAGE

Two-way tables meet Punnett squares

The same mathematics connects family observations to a model of inheritance.

1 / START WITH A TWO-WAY TABLE OF OBSERVATIONS

A cell combines two categories; a margin collects a row or column

An invented sample of 100 parent–child pairs, using only brown and blue categories. Choose one pair uniformly at random. Let $A =$ parent blue and $C =$ child blue.

	Child's eyes		
Parent's eyes	Brown	Blue	Total
Brown	36	24	60
Blue	4	36	40
Total	40	60	100

Divide every count by 100 to obtain a joint probability table.

Joint: both the parent and child have blue eyes.

$$\mathbb{P}(A \cap C) = \frac{36}{100} = 0.36.$$

Marginal: ignore the other classification.

$$\mathbb{P}(A) = 0.40, \qquad \mathbb{P}(C) = 0.60.$$

Conditional: restrict to the 40 blue-eyed parents.

$$\mathbb{P}(C \mid A) = \frac{\mathbb{P}(A \cap C)}{\mathbb{P}(A)} = \frac{36}{40} = 0.90.$$

Here $0.36 \neq 0.40 \times 0.60$: these events are not independent for this random draw.

2 / REPLACE OBSERVED COLOURS WITH INHERITED ALLELES

A Punnett square is a joint table

An **allele** is a version of a gene; a child inherits one allele from each parent. In a one-gene **teaching model**, assign brown to BB, Bb and blue to bb: B is dominant over b. Genotype means the allele combination; phenotype means the observed trait.

For $Bb \times Bb$, an egg or sperm carries B or b, each with probability $\frac{1}{2}$. Assume independent gamete draws.

Egg \ sperm	B $(\frac{1}{2})$	b $(\frac{1}{2})$
B $(\frac{1}{2})$	BB $\frac{1}{4}$	Bb $\frac{1}{4}$
b $(\frac{1}{2})$	bB $\frac{1}{4}$	bb $\frac{1}{4}$

Bb and bB are the same genotype. Add their cells:

$$\mathbb{P}(BB) = \tfrac{1}{4}, \quad \mathbb{P}(Bb) = \tfrac{1}{2}, \quad \mathbb{P}(bb) = \tfrac{1}{4}.$$

The model gives $\frac{3}{4}$ brown, $\frac{1}{4}$ blue per child; four births need not produce a $3{:}1$ split.

3 / MAKE THE WEIGHTS EXPLICIT

Multiplication needs independence

Fix the parents' genotypes. Let p and q be the respective chances that the egg and sperm carry B. With independent draws:

	B (q)	b $(1-q)$
B (p)	pq	$p(1-q)$
b $(1-p)$	$(1-p)q$	$(1-p)(1-q)$

Cells are products. Row sums are $p, 1-p$; column sums are $q, 1-q$. All four cells sum to 1.

$$\mathbb{P}(\text{blue}) \;=\; (1-p)(1-q)$$

For $BB \times bb$ instead, $p = 1, q = 0$: every child is Bb, so the blue probability is 0.

Brown eyes alone do not distinguish BB from Bb. Genotypes set the weights.

4 / WHERE A FAMILIAR CHART GETS ITS FRACTIONS

Why 75%, 18.75% and 6.25%?

A **two-gene toy model** uses two imaginary gene positions (loci). Both parents are $BbGg$; assume the loci assort independently and assign colours by these rules:

Rule	Probability	Colour
At least one B	$\frac{3}{4} = 75\%$	brown
bb, at least one G	$\frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$	green
$bbgg$	$\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$	blue

The last two fractions are 18.75% and 6.25%. The 16 equally likely cells of a 4×4 Punnett square have been grouped into 12, 3 and 1.

These percentages assume this cross and these colour rules. Other brown-eyed parents can have different genotypes and different odds.

5 / KEEP ENOUGH BIOLOGY TO KNOW THE MODEL'S LIMITS

Real eye colour is polygenic

Eye colour depends largely on **melanin** in the iris. Brown eyes have more in its front layers; blue eyes have less. Colours vary continuously, often grouped as brown, green/hazel and blue.

OCA2 affects melanin; a region within **HERC2** regulates OCA2 activity. Other genes contribute too. OCA2 and HERC2 are neighbours on chromosome 15; nearby variants can be inherited together (**linkage**). Independent colour switches are an inadequate model.

Visible colour does not identify all these variants. Predictions need more genetic information and a model tested against data. Even two blue-eyed parents can occasionally have a brown-eyed child.

The mathematician's question: what are the outcomes, what are their weights, and which independence assumptions justify multiplying them?

One framework. Counts divided by a total give empirical probabilities; Punnett cells give model probabilities. Sum cells for combined outcomes; divide by the relevant margin to condition.

Biology sources. MedlinePlus Genetics: Is eye color determined by genetics?; NHGRI: Mendelian inheritance; NHGRI: Linkage.

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